UC16-ECRMDP

Pentaho Data Integration (PDI) Metadata Repository Documentation

Project repository design • ETL organization • Metadata inventory

Prepared as a proposed documentation structure for the project.

1. Purpose and Scope

This document describes a suggested way to organize and document the Pentaho Data Integration assets for the UC16-ECRMDP data-engineering project. It is a design/documentation artifact, not a Pentaho repository export and not proof that a database-backed Pentaho repository has already been configured.

The project context provided indicates a Bronze/Silver/Gold workflow, multiple source datasets (10 datasets were mentioned), and PostgreSQL as the target database named DA-1_Project. Confirm these details against the final project implementation before submission.

2. Repository Organization (Proposed)

```
UC16-ECRMDP
■■■ 00_Database_Connections
■   ■■■ DA-1_Project_PostgreSQL
■■■ 01_Bronze
■   ■■■ BR_Extract_Datasets
■   ■■■ BR_Load_Raw_Data
■■■ 02_Silver
■   ■■■ SL_Clean_Dataset_01 ... SL_Clean_Dataset_10
■   ■■■ SL_Clean_All_Datasets
■■■ 03_Gold
■   ■■■ GL_Create_Fact_Tables
■   ■■■ GL_Create_Dimension_Tables
■   ■■■ GL_Load_Analytics_Tables
■■■ 04_Jobs
■   ■■■ JOB_Run_Bronze_Silver_Gold
■■■ 05_Validation
    ■■■ Check_Null_Values
    ■■■ Check_Duplicate_Keys
    ■■■ Check_Table_Counts
```

Naming convention: use BR_ / SL_ / GL_ prefixes for layer-specific transformations and JOB_ for orchestration jobs. Names above are illustrative; replace them with the actual filenames and object names used in Spoon.

3. Metadata Inventory

Metadata category	What to record	Project example / note
Project	Project name and purpose	UC16-ECRMDP; confirm full expansion/title
Sources	Source file names, formats, paths, and refresh method	Record the actual 10 datasets and source locations
Bronze layer	Raw ingestion transformation, target tables, load date/audit fields	BR_* objects (use actual names)
Silver layer	Cleaning rules, type conversions, null/duplicate handling	SL_* objects; document each rule applied
Gold layer	Business-ready tables, fact/dimension design, keys	GL_* objects; list actual tables and relationships
Database connection	Connection name, DBMS, host, port, database, schema	PostgreSQL / DA-1_Project; do not include passwords
Transformations (.ktr)	Name, purpose, input, output, steps, dependencies	Inventory each actual transformation
Jobs (.kjb)	Name, sequence, triggers, success/failure handling	Document actual orchestration

Metadata category	What to record	Project example / note
Validation	Row counts, null checks, duplicate checks, reconciliation	Include actual validation queries/results

4. ETL Workflow (Conceptual)

Source datasets → Bronze (raw ingestion) → Silver (cleaning and standardization) → Gold (analytics-ready tables) → validation and reporting.

A Pentaho job may orchestrate the sequence of transformations. The exact steps, execution order, and error handling must match the implementation.

5. Repository vs. Target Database

Component	Role
Pentaho repository (if configured)	Stores/organizes PDI jobs, transformations, and repository metadata.
PostgreSQL DA-1_Project	Stores project data tables loaded by the ETL pipeline.
PDF documentation	Explains the repository structure and metadata inventory; it does not execute ETL.
SQL scripts	Create/alter target tables and perform data checks; they are not themselves a Pentaho repository.

6. Implementation Checklist

■ Confirm whether Spoon used a database repository, a file-based repository, or local .ktr/.kjb files.

■ Replace illustrative folder, job, and transformation names with actual project names.

■ List all source datasets and actual Bronze/Silver/Gold target tables.

■ Record connection properties without credentials or secrets.

■ Attach relevant screenshots of Spoon repository tree and successful job execution, if required by the review.

■ Include SQL DDL and validation scripts separately if requested.

7. Accuracy Note

This PDF is a proposed repository documentation template based on project details mentioned in the conversation. It does not contain the actual Pentaho repository metadata, connection credentials, exact transformation names, or exported repository objects. Verify and update it using your Spoon project before treating it as final submission evidence.